



Figure 5: Visualizing the effect of conditioning the label information on the generated samples of the first task (classes 0,1) of split MNIST.



Figure 4: Visualizing the effect of conditioning the label information on the latent representation using random samples from the test data of the first task (classes 0,1) of split MNIST.



Figure 6: Comparison of the effect of providing the z representation to the classifier. ACC and BWT are reported on split MNIST using a different number of replayed samples for each task.

invariant representation using a conditional variational autoencoder (cVAE). This representation is jointly used with specific representation for classification. Pseudo-rehearsal is used to maintain previously learned knowledge. We consider the class incremental learning scenario in which the task identity is not available during inference. The proposed method is evaluated on two standard benchmarks for CL: split MNIST and split Fashion MNIST. Experimental results show the effectiveness of our method and its robustness towards forgetting. Our proposed method outperforms the regularization methods by a large margin. It also achieves a better performance than the state-of-the-art pseudo-rehearsal method by 4% and 3% on split MNIST and split Fashion MNIST respectively. It also reduces the negative backward transfer by 5% and 6% on the two benchmarks respectively. Moreover, we show that shared learned representation by the cVAE helps in mitigate forgetting especially when the number of replayed samples from previous tasks is small. In the future, it would be interesting to analyze the effect of adding more constraints to guarantee that the shared representation does not appear in the class-specific representation. Another



Figure 7: Comparison of the effect of providing the z representation to the classifier. ACC and BWT are reported on split Fashion MNIST using a different number of replayed samples for each task.

interesting extension to the work would be to generate a different number of samples for each task based on how much learning a new task affects its performance. The conditional generation module of our proposed method gives the facility to control the number of generated images for a specific

task.

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